Lecture 13

Tuesday, 28 February 2023 10:32 AM

Thursday MT → 2 questions, open book?

TST for gas-solid reactions

Eley-Rideal : $A_{(g)} + B^* \overset{TS}{\rightleftharpoons} C^*$

Assuming TS is close to surface (adsorbed)

$$\Rightarrow K^* = \frac{a_{TS^*}}{a_A\, a_{B^*}} = \frac{\theta_{TS}/\theta_*}{\frac{p_A}{p^*}\frac{\theta_B}{\theta_*}} = \frac{\theta_{TS}}{\frac{p_A}{p^*}\theta_B}$$

$$\theta_{TS} = K^* \theta_B \frac{p_A}{p^*} \qquad \text{(Only 1 adsorbed species)}$$

TST assumption (rate prop. to coverage of TS)

$$r = \nu \theta_{TS} = \nu K^{\ddagger} \theta_B \frac{p_A}{p^*}$$

$$\boxed{r = \frac{k_B T}{h} K^{\ddagger} \theta_B \frac{p_A}{p^\circ}}$$

p°: No reaction coord.

$$K^{\ddagger} = e^{\frac{-\Delta\mu^{\circ\ddagger}}{k_B T}} \quad \text{where} \quad \Delta\mu^{\circ\ddagger} = \mu^{\circ\ddagger} - \mu^\circ_{A_{(g)}} - \mu^\circ_{B^*}$$

(Adsorbed TS) $\mu^{\circ\ddagger} = -k_B T \ln q^\circ_{vib,TS} \cdot e^{-\beta(E_{DFT_{TS}} + ZPE_{TS})}$

$$\mu^\circ_{B^*} = -k_B T \ln q^\circ_{vib,B^*} \cdot e^{-\beta(E_{DFT_{B^*}} + ZPE_{B^*})}$$

$$\mu^\circ_A = -k_B T \frac{\partial}{\partial N_{A_{(g)}}} \left[\ln \left\{ Q_{trans,A}\, q^{N_A}_{vib,A}\, q^{N_A}_{rot,A}\, e^{-\beta N_A (E_{DFT_A} + ZPE_A)} \right\} \right]$$

$$= -k_B T \ln q^\circ_{vib,A} - k_B T \ln q^\circ_{rot,A} - k_B T \frac{\partial}{\partial N_A}\left(\ln Q^\circ_{trans,A}\right) + E_{DFT_A} + ZPE_A$$

$$\therefore K^{\ddagger} = \frac{q_{vib,TS^*}}{q_{vib,B^*}\, q^\circ_{vib,A}\, q^\circ_{rot,A}\, e^{\frac{\partial \ln Q^\circ_{trans,A}}{\partial N_A}}} \cdot e^{-\beta(\Delta E_{DFT} + \Delta E_{ZPE})}$$

ΔE: activation energy

$A_1, A_2 \Longleftrightarrow A_2 A_1$ (Same?)

$$Q^\circ_{trans} = \frac{q^{\circ N}_{trans}}{N!}$$

$$e^{\frac{\partial \ln Q^\circ_{trans,A}}{\partial N_A}} = \left(\frac{q_{trans,A}}{N}\right)'$$

$$\boxed{r = \frac{k_B T}{h} \frac{q^\circ_{vib,TS^*}}{q_{vib,B^*}\, q^\circ_{vib,A}\, q^\circ_{rot,A} \left(\frac{q^\circ_{trans,A}}{N}\right)} e^{-\beta(\Delta E + \Delta E_{ZPE})} \frac{p_A}{p^\circ}\theta_B}$$

Note:

1) All q_{vib}'s have ZPE part removed
2) q° for vib/rot partition functions is that of isolated species (gas or surface)
3) p° is often lumped into rate constant (careful of units)
4) $q_{electronic}$ is part of the E_{DFT} expression

Alternate TST w/o Statistical Mechanics

In general, $k = \frac{k_B T}{h} e^{\frac{-\Delta\mu^{\ddagger}}{k_B T}}$ ← No rxn coord.

$$\Delta\mu^{\circ\ddagger} = \Delta\left(\frac{\partial F}{\partial N_i}\right)^{\circ\ddagger} = \Delta \frac{\partial}{\partial N_i}\left(U_i - TS_i\right) \qquad (H_i)$$

$$= \Delta\bar{U}^{\circ\ddagger} - T\Delta\bar{S}^{\circ\ddagger}$$

$\bar{U}, \bar{S}$: partial molar quantities

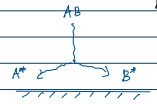
$$\boxed{k = \frac{k_B T}{h} e^{\frac{\bar{S}^{\circ\ddagger}}{k_B}} \cdot e^{-\frac{\bar{U}^{\circ\ddagger}}{k_B T}}}$$

For pure species (i.e. @ partial P_i), or non-interacting surface-species, partial molar is same as "normal" molar quantity

Many ways to calc. $\bar{S}^\circ$:
+ experimental values
+ Stat. mech.
+ Correlations (for surface species)
+ $\Delta S^{\circ\ddagger} \approx 0$ (if reaction is entirely on the surface)

Ex: AB ? Dissociative adsorption

Ex: Dissociative adsorption



a) Simple assumption: $S^\circ_{TS} \approx 0$, thus $\Delta S^{\circ\ddagger} = S^\circ_{TS} - S^\circ_{AB(g)} \approx -S^\circ_{AB(g)}$

and $k = \frac{k_B T}{h p^\circ} e^{\frac{-S^\circ_{AB}}{k_B}} e^{\frac{-\Delta U^{\circ\ddagger}}{k_B T}}$ ← thermo tables

Collision Theory (rates of adsorption/desorption of ideal gases w/o dissociation)

Kinetic Theory of Gases: Flux of molecule hitting a surface $= \frac{P_i}{\sqrt{2\pi m k_B T}}$ [?] $\frac{1}{m^2 \cdot s}$

On a per site basis, $\boxed{k^\circ_{ads} = \frac{1}{\sqrt{2\pi m k_B T} \cdot \rho}}$; $\rho \equiv \frac{\text{sites}}{\text{area}}$ (or n_s)

↳ If adsorption is activated, add a Boltzmann factor and scale by number of vacant sites:

$$k = k^\circ_{ads} e^{\frac{-\Delta E^{act}}{k_B T}}$$

$$r_{ads} = k P_i \theta^n_{*} \quad ; \quad n = 1 \text{ or } 2$$

($n=1$: simple adsorption; $n=2$: dissociative adsorption)

↳ Same expression from TST: assume TS is a 2D ideal gas

+ Calculate rate from 3D to 2D gas

Partition functions: $Q_{2D} = \frac{q^N_{2D}}{N!} = \left(\frac{2\pi m k_B T}{h^2} A\right)^N \frac{1}{N!}$; $\ln Q_{2D} = N \ln\left[\frac{2\pi m k_B T}{h^2} A\right] - N \ln N + N$

$$Q_{3D} = \frac{q^N_{3D}}{N!} = \left[\left(\frac{2\pi m k_B T}{h^2}\right)^{3/2} V\right]^N \frac{1}{N!} \quad ; \quad \ln Q_{3D} = N \ln\left[\left(\frac{2\pi m k_B T}{h^2}\right)^{3/2} V\right] - N \ln N + N$$

$$F_{2D} = -k_B T \ln Q_{2D} \quad ; \quad \mu^{(TS)}_{2D} = \frac{\partial F_{2D}}{\partial N} = -k_B T \left[\ln\left(\frac{2\pi m k_B T}{h^2} A\right) - \ln N\right]$$

$$F_{3D} = -k_B T \ln Q_{3D} \quad ; \quad \mu_{3D} = \frac{\partial F_{3D}}{\partial N} = -k_B T \left[\ln\left(\frac{2\pi m k_B T}{h^2}\right)^{3/2} V - \ln N\right]$$

@ quasi-equil, $\mu^{TS}_{2D} = \mu_{3D}$

$$\frac{2\pi m k_B T}{h^2} \frac{A}{N_{2D}} = \left(\frac{2\pi m k_B T}{h^2}\right)^{3/2} \frac{V}{N_{3D}}$$

$$\left(\frac{N}{A}\right)_{2D} = \frac{P}{k_B T} \sqrt{\frac{h^2}{2\pi m k_B T}}$$

From TST,

$$r = \gamma \left(\frac{N}{A}\right)_{2D} = \frac{k_B T}{h} \left(\frac{N}{A}\right)_{2D} = \frac{P}{\sqrt{2\pi m k_B T}}$$

So, $\boxed{k^\circ_{ads} = \frac{1}{\sqrt{2\pi m k_B T} \cdot \rho}}$ ← Same as collision theory

Thermodynamic consistency of rate constants

Seek to relate kinetic and thermo parameters for elementary reactions

Ex: $A(g) + B^* \rightleftharpoons C^*$

@ equil., $K = \frac{a_{C^*}}{a_A a_{B^*}} = \frac{\theta_C}{\frac{P_A}{P^\circ} \theta_B}$

& $r_f = r_r$ ($r_{net} = 0$)

$$k_f P_A \theta_* = k_r \theta_c \Rightarrow \frac{k_f}{k_r} = \frac{\theta_c}{P_A \theta_B}$$

$$\therefore \frac{\theta_c}{\frac{P_A}{P^\circ}\theta_B} = \frac{k_f P^\circ}{k_r} = K$$

$$\therefore \boxed{k_r = \frac{k_f P^\circ}{K}}$$

Microkinetics

Overview (Langmuir-Hinshelwood):

Consider reaction mechanism:

1) $A_2(g) + 2* \rightleftharpoons 2A^*$ } might be QE
2) $B_2(g) + 2* \rightleftharpoons 2B^*$ }
3) $A^* + B^* \rightleftharpoons AB^* + *$
4) $AB^* + B^* \rightleftharpoons AB_2^* + *$
5) $AB_{(g)} + * \rightleftharpoons AB^*$ } Simple adsorption
6) $AB_{2(g)} + * \rightleftharpoons AB_2^*$ }

Make balances for each adsorbed species: (rates must be $[\text{time}]^{-1}$ (turn over frequency))

$$A^*: \frac{d\theta_A}{dt} = 2r_1 - r_3$$

$$B^*: \frac{d\theta_B}{dt} = 2r_2 - r_3 - r_4$$

$$AB^*: \frac{d\theta_{AB}}{dt} = r_3 - r_4 + r_5$$

$$AB_2^*: \frac{d\theta_{AB_2}}{dt} = r_4 + r_6$$

r's are net rates (can be <0)

@ SS: $\frac{d\theta_i}{dt} = 0 \Rightarrow$

$2r_1 - r_5 = 0$
$2r_2 - r_3 - r_4 = 0$
$r_3 - r_4 + r_5 = 0$
$r_4 + r_6 = 0$
} express in terms of product rates, r_5 and r_6

$\Downarrow$

$r_4 = -r_6 \qquad r_2 = -\frac{1}{2}r_5 - r_6$
$r_3 = -r_5 - r_6 \qquad r_1 = -\frac{1}{2}r_5 - \frac{1}{2}r_6$

} Two linearly independent rates [6 unknowns, 4 equations]

$\Downarrow$

Need additional info (Use TST-like rate expressions)

$$r_1 = k_1 P_{A_2} \theta_*^2 - k_{-1} \theta_A^2$$

$$r_2 = k_2 P_{B_2} \theta_*^2 - k_{-2} \theta_B^2$$

$$r_3 = k_3 \theta_A \theta_B - k_{-3} \theta_{AB} \theta_*$$

$$r_4 = k_4 \theta_{AB} \theta_B - k_{-4} \theta_{AB_2} \theta_*$$

$$r_5 = k_5 P_{AB} \theta_* - k_{-5} \theta_{AB_2}$$

Lecture 14

If k's and P_i's are known, we have 5 variables: $\theta_A, \theta_B, \theta_{AB}, \theta_{AB_2}, \theta_*$ } 5 eqs, 5 unknowns

Lastly, $\theta_* = 1 - \theta_A - \theta_B - \theta_{AB} - \theta_{AB_2}$

If k_i's and P_i's are known, we have 5 variables: $\theta_A, \theta_B, \theta_{AB}, \theta_{AB_2}, \theta_*$

4 eq's + $\theta_* = 1 - \theta_A - \theta_B - \theta_{AB} - \theta_{AB_2}$ } 5 eqs, 5 unknowns

Comments:

+ solution of eqns for θ_i's gives rigorous rate. Can always do this
+ Assumption: L-H kinetics is valid

More practical: Assume some steps are rate limiting, and some are quasi-equilibrated

↳ ex: Assume 1, 2, 5, 6 are QE → why choose 4? → simple adsorption are often QE

$$K_1 = \frac{\theta_A^2}{\frac{P_A}{P^\circ}\theta_*^2} \Rightarrow \theta_A = \sqrt{K_1 \cdot \frac{P_{A_2}}{P_0}}\,\theta_*$$

$$\theta_B = \sqrt{K_2 \frac{P_{B_2}}{P^\circ}}\,\theta_* \quad , \quad \theta_{AB} = K_5 \frac{P_{AB}}{P^\circ}\theta_* \quad ; \quad \theta_{AB_2} = K_6 \frac{P_{AB_2}}{P^\circ}\theta_*$$

(Expressing everything in terms of θ_*)

From using site balance:

$$\theta_* = \frac{1}{1 + \sqrt{K_1 \frac{P_{A_2}}{P^\circ}} + \sqrt{K_2 \frac{P_{B_2}}{P^\circ}} + K_5 \frac{P_{AB}}{P^\circ} + K_6 \frac{P_{AB_2}}{P^\circ}}$$

(A^*) (B^*) (AB^*) (AB_2^*)

Finally, substitute into r_i's that are not QE:

$$r_3 = k_3 \sqrt{K_1 \frac{P_{A_2}}{P^\circ}} \sqrt{K_2 \frac{P_{B_2}}{P^\circ}}\,\theta_*^2 - k_{-3} K_3 \frac{P_{AB}}{P^\circ}\theta_*^2$$

$$r_4 = k_4 \sqrt{K_2 \frac{P_{A_2}}{P^\circ}}\, K_5 \frac{P_{AB}}{P^\circ}\theta_*^2 - k_{-4} K_6 \frac{P_{AB_2}}{P^\circ}\theta_*^2$$

Product production rates:

$$r_{AB_{(g)}} = -r_5 = r_3 - r_4$$

$$r_{AB_{2(g)}} = -r_6 = r_4$$

Explore simpler case in more detail:

1) $A_{(g)} + * \rightleftharpoons A^*$

2) $A^* \rightleftharpoons B^*$

3) $B_{(g)} + * \rightleftharpoons B^*$

$$\frac{d\theta_A}{dt} = 0 = r_1 - r_2 \quad ; \quad \frac{d\theta_B}{dt} = 0 = r_2 + r_3$$

$$r_1 = k_1 P_A \theta_* - k_{-1}\theta_A$$

$$r_2 = k_2 \theta_A - k_{-2}\theta_B \quad ; \quad \theta_* = 1 - \theta_A - \theta_B$$

$$r_2 = k_3 P_B \theta_* - k_{-3}\theta_B$$

Case 1: (1) & (2) are QE

$$\theta_A = k_1 \frac{P_A}{P^\circ}\theta_* \quad ; \quad \theta_B = K_3 \frac{P_B}{P^\circ}\theta_*$$

$$\theta_* = \frac{1}{1 + K_1 \frac{P_A}{P^\circ} + K_3 \frac{P_B}{P^\circ}}$$

$$r_{B(g)} = -r_3 = r_3 = k_3 K_1 \frac{P_A}{P^\circ}\theta_* - k_{-3} K_3 \frac{P_0}{P^\circ}\theta_*$$

Case 2: (1) is QE

$\theta_A = K_1 \frac{P_A}{P^\circ} \theta_*$; Find θ_B from stoichiometric SS eqn:

$0 = r_2 + r_3$

$k_2 \theta_A - k_{-2} \theta_B = k_{-3} \theta_B - k_3 P_B \theta_*$

$\Downarrow$

$\theta_B = \frac{k_2 K_1 \frac{P_A}{P^\circ} + k_3 P_B}{k_{-2} + k_{-3}} \theta_*$

$\theta_* = 1 - \theta_A - \theta_B$

$r_{avg} = -r_3 = r_2$ (either works here) ← can use any rate eq since neither are QE

Case 3: Only (3) is QE (exercise)

Case 4: No steps are QE (No assumption about kinetics apart from Langmuir Hinshelwood)

$r_1 = r_2 \Rightarrow k_1 P_A \theta_* - k_{-1} \theta_A = k_2 \theta_A - k_{-2} \theta_B$

$r_2 = -r_3 \Rightarrow k_2 \theta_A - k_{-2} \theta_B = k_{-3} \theta_B - k_3 P_B \theta_*$

$\theta_A + \theta_B + \theta_* = 1$

} SS balance to get θ_A, θ_B

$\theta_A = \frac{k_1 P_A \theta_* + k_{-2} \theta_B}{\underbrace{k_{-1} + k_2}_{\Delta k_a}}$; $\theta_B = \frac{k_2 \theta_A + k_3 P_B \theta_*}{\underbrace{k_{-2} + k_{-3}}_{\Delta k_b}}$

$\theta_B = \frac{k_2}{\Delta k_b} \left[\frac{k_1}{\Delta k_a} P_A \theta_* + \frac{k_{-1}}{\Delta k_a} \theta_B \right] + \frac{k_3}{\Delta k_b} P_B \theta_*$; $\theta_B = \frac{\frac{k_1 k_2}{\Delta k_a \Delta k_b} P_A + \frac{k_3}{\Delta k_b} P_B}{1 - \frac{k_2 k_{-2}}{\Delta k_a \Delta k_b}} \theta_*$

$\theta_A = \left[\frac{\frac{k_1 k_2 k_{-2}}{\Delta k_a^2 \Delta k_b} P_A + \frac{k_2 k_3}{\Delta k_a \Delta k_b}}{1 - \frac{k_2 k_{-2}}{\Delta k_a \Delta k_b}} + \frac{k_1}{\Delta k_a} P_A \right] \theta_*$

By taking various asymptotic limits of k_i's, we can explore the meaning of QE

Non QE = RLS
- Can have multiple RLS
- Each step can have different degree of RLS (Campbell score)
- Campbell
  - 0.0001 → QE
  - 1 → RLS

Q2 on HW